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Research paper

# A study on crude oil price forecasting model integrating CEEMDAN-VMD multiscale decomposition with CNN-BiLSTM

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ABSTRACT

As a key global energy benchmark, WTI crude oil prices exhibit significant volatility driven by a complex interplay of economic, geopolitical, and financial factors. This results in highly nonlinear and non-stationary price behavior, posing substantial challenges to accurate forecasting. This study proposes a hybrid forecasting model (CEEMDAN-VMD-CNN-BiLSTM) to improve the accuracy of WTI crude oil price predictions, addressing its volatility driven by complex economic, geopolitical, and financial factors. The model integrates Complementary Ensemble Empirical Mode Decomposition (CEEMDAN), Variational Mode Decomposition (VMD), Convolutional Neural Networks (CNN), and Bidirectional Long Short-Term Memory (BiLSTM). Pearson correlation analysis selects the top four indicators from nine potential features, including LBMA gold price, Brent crude oil price, USD/CNY exchange rate, and Federal Funds rate. The raw price data and features undergo dual decomposition, with CNN extracting local temporal features and BiLSTM capturing bidirectional dependencies. The model, trained on data from 1997–2017 and validated on 2017–2025, achieves a MAPE of 3.66 % and an $R^2$ of 95.94 %, outperforming BiLSTM and CNN-BiLSTM. Comparative analysis with eight other models, including LSTM, SVM, and Transformer, confirms the superior performance of the proposed model, with a MAPE of 5.57 % and an $R^2$ of 93.87 %. This approach enhances forecasting accuracy and provides insights for optimizing crude oil futures markets and risk management strategies.

## 1. Introduction

Fluctuations in WTI crude oil prices have profound implications for the global economy. As a key pricing benchmark in international markets, WTI reflects both domestic and global supply-demand dynamics [1–3], and establishes a close linkage with Brent crude oil prices through the futures market [4,5]. In recent years, WTI prices have experienced sharp volatility driven by multiple factors such as geopolitical conflicts, energy transition policies, and speculative activities in financial markets [6,7]. For example, in 2022, supply shortages caused by the Russia–Ukraine conflict pushed prices up to $120 per barrel, while in 2023, expectations of a global economic slowdown led to a retreat to the $70–80 range. These price movements exhibit complex nonlinear and cross-cycle transmission mechanisms influenced by geopolitical risks, monetary policy changes, market sentiment, and energy substitution effects—dynamics that go beyond the explanatory scope of traditional supply-demand models. Therefore, the motivation of this study is to construct a multidimensional forecasting framework that integrates macroeconomic indicators and market dynamics, aiming to improve the accuracy and robustness of crude oil price forecasting in an environment characterized by high uncertainty.

Forecasting crude oil prices remains a critical focus in energy economics [8], leading to the development of diverse models based on various data characteristics and theoretical assumptions. Liang Q et al. [9] integrated GRU with nonlinear Granger causality and group regularization to enhance accuracy using multi-dimensional factors. Chen Y et al. [10] built a hybrid model combining GRU, ARIMA, LSTM, and XGBoost under the HMOFTTA framework to improve forecasting stability. Liu J et al. [11] applied complex networks and multi-source sentiment analysis to forecast interval-valued prices. Jha N et al. [12]

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used SVR alongside ridge regression and LASSO to improve accuracy and robustness. Tan J et al. [13] proposed a multi-scale decomposition framework to capture cross-scale patterns. Liu L et al. [14] introduced the JWH model with time-varying weights and Shannon entropy to enhance predictions. Du X et al. [15] developed an OVMD (V–P)-DRL-Ensemble model leveraging time-frequency features and deep reinforcement learning. Zhao Y et al. [16] used Granger causality to build an impulse-response network linking oil price shocks to national indices. Zhang W et al. [17] analyzed aviation fuel and diesel markets using nonlinear time series methods. Wang S et al. [18] employed a multi-level nested copula model to study geopolitical influences on oil security. Lee J et al. [19] compared deep learning models like MLP, LSTM, GRU, and TCN using futures price data, noting reduced accuracy for longer maturities. Finally, Jahandoost A [20] proposed the K-means-DSD-LSTM model, which combines clustering and deep learning to address volatility and overfitting, offering superior predictive performance.

Forecasting crude oil prices, a critical issue in energy economics, has drawn wide scholarly interest [20]. With advancements in machine learning and deep learning, numerous models have emerged based on diverse theoretical assumptions and data characteristics. Liang Q et al. [9] proposed a GRU-GC model that combines Gated Recurrent Units with nonlinear Granger causality and group regularization to improve accuracy using multidimensional inputs like OPEC+ output, GDP, and the VIX index. Chen Y et al. [10] built a hybrid model incorporating GRU, Granger causality, and HMOFTTA, integrating ARIMA, LSTM, and XGBoost to enhance forecasting stability. Liu J et al. [11] utilized complex networks and multi-source data, including news sentiment, for interval-valued price prediction. Jha N et al. [12] applied SVR alongside ridge regression and LASSO, demonstrating superior accuracy and robustness. Tan J et al. [13] developed a multi-scale decomposition framework to model nonlinear, cross-scale patterns. Liu L et al. [14] introduced a Jaynes Weighting Hybrid (JWH) model with time-varying weights and Shannon entropy to improve accuracy. Du X et al. [15] proposed an OVMD(V–P)-DRL-Ensemble model leveraging deep reinforcement learning and volume-price features. Zhao Y et al. [16] used Granger causality and impulse-response networks to assess oil price shocks on national indices. Zhang W et al. [17] explored aviation fuel and diesel markets using nonlinear time series analysis. Wang S et al. [18] applied a nested copula model to examine the influence of geopolitical risks on oil security. Lee J et al. [19] employed deep learning models (MLP, RNN, LSTM, GRU, CNN, TCN) using futures prices, though accuracy diminished with longer maturities. Jahandoost A [20] introduced the K-means-DSD-LSTM model, combining clustering and deep learning to address price volatility and overfitting, achieving improved performance over traditional methods.

Recent studies have significantly enhanced the accuracy and robustness of crude oil and natural gas price forecasting by applying multimodal decomposition techniques. He H et al. [21] combined Variational Mode Decomposition (VMD) with machine learning to extract high- and low-frequency components for capturing short- and long-term price trends. Yuan J et al. [22] proposed a clustered regular incremental monotonic weighting strategy, integrating models like ARIMA, SVM, ELM, LSTM, XGBoost, and Light-GBM to optimize forecast balance and robustness. Wang J et al. [23] introduced an optimized VMD (OVMD) method and developed a hybrid model, OVMD-ATT-SSF-DBILSTM, validated by multiple evaluation metrics. Sun B et al. [24] used Ensemble Empirical Mode Decomposition (EEMD) to smooth time series data and applied a trivariate clustering algorithm for price range prediction. Yang R et al. [25] built an ensemble model incorporating model ranking, multi-objective learning, and residual correction, using VIKOR, TOPSIS, and GRA to select optimal solutions validated with WTI and Brent crude oil data.

Table 1 provides a comprehensive survey of recent crude oil price forecasting models proposed by various scholars. It showcases a wide array of machine learning and hybrid approaches—ranging from traditional models like ARIMA and SVM to advanced deep learning architectures such as LSTM, CNN, and GRU, as well as hybrid frameworks like XGBoost-LSTM and EEMD-CNN-BiLSTM-QR. The forecasting targets include major crude oil benchmarks such as WTI and Brent, as well as regional oils and natural gas. Model accuracy varies significantly across studies, with some achieving over 99% prediction accuracy, while others report moderate performance or use error metrics like RMSE and MAPE. This diversity highlights both the complexity of oil price prediction and the growing role of intelligent algorithms in this domain.

**Table 1**
Survey of crude oil price forecasting models proposed by various scholars.

| Reference | Model | Accuracy | Forecasting Target |
|---|---|---|---|
| Göncü A, Kuzubaş T U [26] | XGBoost, CNN, Logistic, ANN, SVC, RF, XTrees | 56.65 %, 43.28 %, 46.56 %, 44.65 %, 46.94 %, 44 %, 45 % | Brent Crude Oil |
| Kaymak Ö Ö [27] | Fuzzy Time Series, ANN, SVM | — | WTI Crude Oil |
| He H [21] | VMD-ML | 61.4 %, 67.4 %, 65.1 % | WTI Crude Oil |
| Cen Z [28] | EEMD-LSTM | — | WTI Crude Oil, Brent Crude Oil |
| Busari G A [29] | AdaBoost-GRU | 98.36 %, 96.98 %, 96.46 %, 97.53 % | WTI Crude Oil |
| Bristone M [30] | LSTM | 97.09 % | WTI, UK Brent, UAE Dubai, Russia Urals, Qatar Dukhan, Angola Zarzaitine, Libya Brega, Iran Light, Egypt Suez B, Norway Ekofisk |
| Simsek Ahmed Ihsan [31] | XGBoost-LSTM | 99.97 %, 99.91 %, 99.87 % | WTI Crude Oil |
| He M [32] | LSTM | 98.42 %, 98.82 %, 99.76 % | WTI Crude Oil |
| Jha Nimish [12] | SVM-RBF | 99.51 %, 92.46 % | WTI Crude Oil |
| Asl M G [33] | LSTM, SNN, RNN | RMSE: 1.6342, 9.976, 3.2402, 16.24 | Iranian Crude Oil and Gas |
| Jin B [34] | RT, SVR, AR, AR-GARCH | 91 % | WTI Crude Oil, Brent Crude Oil, New York Harbor No 2 Heavy Oil, Henry Hub Natural Gas |
| Oyewola D O [35] | DLQL, DLAQL, LSTM | 47.24 %, 43.74 %, 39.76 % | Liquefied Natural Gas |
| Shen L [36] | EEMD-CNN-BiLSTM-QR | RMSE: 1.4278, 1.3954, MAPE: 1.7517, 1.8661 | WTI Crude Oil, Brent Crude Oil |
| Karasu S [37] | Logistic Chaotic Map | 99.266 %, 99.158 %, 98.795 %, 98.908 % | WTI Crude Oil, Brent Crude Oil |
| Naderi M [38] | LSSVM, GP, ANN, ARIMA, BA | 99.6 %, 96.11 %, 97.8 % | Oil Prices, Interest Rates, Tina Gas |
| Wang Y [39] | Kernel Regularized Non-Homogeneous Grey Riccati Model | MAPE: 0.4074 %, 0.4205 %, 0.6395 %, 0.4862 % | Oil Reserves |
| Lahmiri S [40] | GRP, SVR, Rtrees, KNN, Bayesian | RMSE: 0.0416, 0.0253 | Brent Crude Oil, WTI Crude Oil |

Various scholars have made significant strides in forecasting crude oil prices by employing different models, each with its unique strengths and limitations. Some models, such as Liang Q et al. [9], Chen Y et al. [10], and Liu J et al. [11], excel in incorporating multiple data sources and complex relationships, enhancing prediction accuracy and stability. However, these models often suffer from high computational complexity, potential overfitting, and dependency on external data, which can introduce noise. Models like Jha N et al. [12] and Tan J et al. [13] effectively handle non-linear relationships and temporal patterns but may struggle to capture the dynamic interactions of external factors. The OVMD-VMD-DRL-Ensemble strategy proposed by Du X et al. [15] leverages deep reinforcement learning, offering improved forecasting, but might lack generalizability across different market conditions. Sun B et al. [24] and Yang R et al. [25] provide solid ensemble approaches, focusing on price ranges or multi-objective solutions, but may not address the complexities of long-term trend forecasting as effectively as models like LSTM or CNN. In summary, various forecasting models for crude oil prices have been developed, each with strengths in different areas, such as incorporating multiple data sources, handling non-linear relationships, and leveraging ensemble techniques. However, these models often face challenges such as high computational complexity, potential overfitting, dependency on external data, difficulty in capturing dynamic interactions, and limitations in generalizability across diverse market conditions. Some models excel at short-term forecasting, while others focus on price ranges or multi-objective solutions but may struggle with accurately predicting long-term trends.

In addition to oil price predictions, researchers are also actively conducting research on global renewable energy and other currency predictions [41–44]. By applying machine learning algorithms such as RF and LSTM, researchers analyze energy data and currency fluctuations from countries around the world and use feature selection techniques to improve the accuracy of predictions. These studies not only support the global sustainable Development Goals, but also help optimize the integration of renewable energy and grasp the future trends of currency.

Despite advancements in the accuracy and stability of crude oil price forecasting models, several limitations persist. These include challenges in addressing complex nonlinear relationships, excessive model complexity due to the interplay of multiple factors, and the influence of various data characteristics on model performance. Many existing models struggle to capture both long-term and short-term dependencies in time-series data, which leads to suboptimal predictions in volatile market conditions. Additionally, while traditional machine learning and deep learning approaches can identify some of the patterns in price fluctuations, they often fail to fully utilize the combined effects of diverse data sources, such as news sentiment and external influences, that play a crucial role in shaping crude oil price dynamics.

In contrast, the proposed CEEMDAN-VMD-CNN-BiLSTM model effectively addresses these challenges by integrating multiple decomposition techniques (CEEMDAN and VMD) with deep learning methods (CNN and BiLSTM). This hybrid approach allows the model to capture both short-term fluctuations and long-term trends by decomposing the crude oil price series into multiple frequency components. The CNN layer enhances the model's ability to extract local features, while the BiLSTM layer captures both forward and backward temporal dependencies, improving forecasting accuracy. By leveraging these advanced techniques, the CEEMDAN-VMD-CNN-BiLSTM model outperforms traditional models in terms of accuracy and robustness, providing more reliable forecasts under volatile market conditions.

Currently, the fluctuations in crude oil prices (WTI) are inherently the result of the interplay between the global energy market, macroeconomic cycles, financial speculation, and geopolitical dynamics. The multidimensional interdependencies within this system evolve dynamically over short, medium, and long-term periods. Therefore, this study utilizes the CEEMDAN-VMD-CNN-BiLSTM model to forecast WTI crude oil prices, incorporating a range of parameters, including WTI crude oil price, Dow Jones Industrial Average, Economic Uncertainty Index, CBOT Volatility Index (VIX), effective Federal Funds rate, Nasdaq 100 Index, Brent crude oil price, Henry Hub natural gas spot price, USD/CNY exchange rate, and LBMA Gold Price [31,45–47].

Research Motivation:

WTI crude oil prices are highly volatile and influenced by a wide range of complex factors. Traditional forecasting models often struggle to capture nonlinear dependencies and long-term trends, especially when incorporating diverse elements such as geopolitical risks, market sentiment, and macroeconomic indicators. This study is driven by the need for a more accurate and robust forecasting approach capable of handling such complexities under high uncertainty.

Research Objective:

This study introduces a novel CEEMDAN-VMD-CNN-BiLSTM hybrid forecasting framework to address the challenges of nonlinear and non-stationary WTI crude oil price data. By integrating CEEMDAN and VMD for multiscale decomposition, CNN for feature extraction, and BiLSTM for temporal modeling, the framework improves forecasting accuracy and robustness. Effective feature selection through Pearson correlation analysis identified four key factors—LBMA Gold Price, Brent Crude Oil Price, USD/CNY Exchange Rate, and Federal Funds Rate—optimizing the model's input and reducing noise. Ablation experiments validated the synergistic effect of integrating these components, leading to a significant performance boost. The CEEMDAN-VMD-CNN-BiLSTM model outperforms eight benchmark models across key metrics, demonstrating its advanced predictive capabilities in crude oil price forecasting.

The primary contributions of this study are as follows:

(1) Development of a Hybrid Forecasting Framework: A novel CEEMDAN-VMD-CNN-BiLSTM framework is proposed to address the nonlinear and non-stationary nature of WTI crude oil price series, improving forecasting accuracy through multiscale feature learning.
(2) Effective Feature Selection: Four key factors—LBMA Gold Price, Brent Crude Oil Price, USD/CNY Exchange Rate, and Federal Funds Rate—were identified through Pearson correlation analysis to optimize model input and reduce noise.
(3) Validation of Synergistic Model Components: Ablation experiments confirmed that integrating CEEMDAN, VMD, CNN, and BiLSTM components significantly improves forecasting performance.
(4) Superior Predictive Performance: The CEEMDAN-VMD-CNN-BiLSTM model outperforms eight benchmark models across key metrics, demonstrating its high accuracy, robustness, and advanced capabilities in crude oil price forecasting.

The subsequent sections of this study are as follows: Section 2 introduces the sources of crude oil prices and related parameter indicators. Section 3 presents the theoretical framework and methodology of the combined crude oil price forecasting model based on CEEMDAN-VMD-CNN-BiLSTM. Sections 3 and 4 provides a detailed comparison of experimental results and discussion. Section 4 concludes the study with key findings.

## 2. Methods

### 2.1. Identification of key factors and variables affecting crude oil prices

The first step in constructing an accurate crude oil price forecasting model is to identify and select key influencing factors that effectively

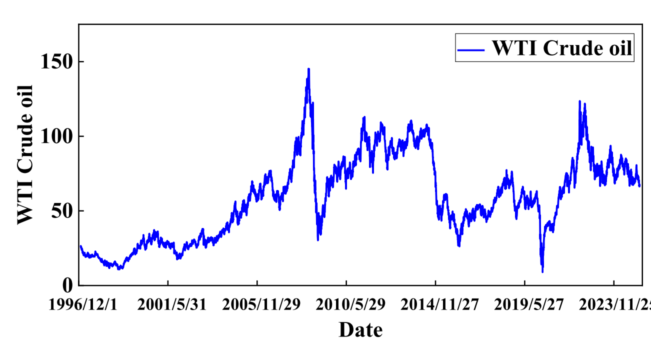


**Fig. 1.** Trends in crude oil prices (WTI).

reflect the complex dynamics of oil price fluctuations. As shown in Fig. 1, WTI (West Texas Intermediate) crude oil, one of the most important global pricing benchmarks, is subject to significant price volatility driven by the interplay of macroeconomic conditions, geopolitical factors, financial market sentiment, and energy supply-demand fundamentals. Therefore, this study initially selects a multivariate feature set consisting of nine potential driving factors, aimed at comprehensively capturing the various forces that influence WTI prices, as illustrated in Fig. 2. These features include: Energy Market: Brent crude oil price, Henry Hub natural gas price; Macroeconomic and Policy Factors: Federal Funds rate, USD/CNY exchange rate, Economic Uncertainty Index; Financial and Sentiment Indicators: Dow Jones Industrial Average, Nasdaq 100 Index, VIX Index, LBMA Gold Price.

The WTI crude oil price is set as the target output variable of the model, which aligns with the practice of numerous existing studies in the field [48–54], confirming its representativeness as a core forecasting target. To ensure the reliability and authority of the data, the data used in this study is sourced from widely recognized and publicly accessible databases. The Economic Uncertainty Index, CBOT Volatility Index (VIX), Federal Funds Effective Rate, Nasdaq 100 Index, Brent Crude Oil Price, Henry Hub Natural Gas Spot Price, and USD/CNY Exchange Rate data were obtained from the Federal Reserve Bank of St. Louis Economic Database (FRED, https://www.stlouisfed.org); the Dow Jones Industrial Average data was sourced from Investing.com (https://cn.investing.com); and the LBMA Gold Price data was retrieved from the World Gold Council China website (https://china.gold.org). This study employs a long-span time series dataset, covering daily observations from January 7, 1997, to March 14, 2025.

To analyze the direct correlation between WTI crude oil prices and nine relevant indicators, this study conducted a Pearson correlation coefficient analysis using daily frequency data. As shown in Fig. 3 and Table 2, the Pearson correlation results reveal that LBMA Gold Price, Brent Crude Oil Price, USD/CNY Exchange Rate, and Federal Funds Effective Rate exhibit strong correlations with WTI crude oil prices, whereas the CBOT Volatility Index, Economic Uncertainty Index, Dow Jones Industrial Average, Nasdaq 100 Index, and Henry Hub Natural Gas Spot Price show weaker correlations. Consequently, the four indicators with the strongest correlations were selected as input features for the experiment, while the WTI crude oil price was used as the output feature. Furthermore, to mitigate the impact of varying magnitudes among the indicators, the data were normalized Figs. 3, 5 and 6.

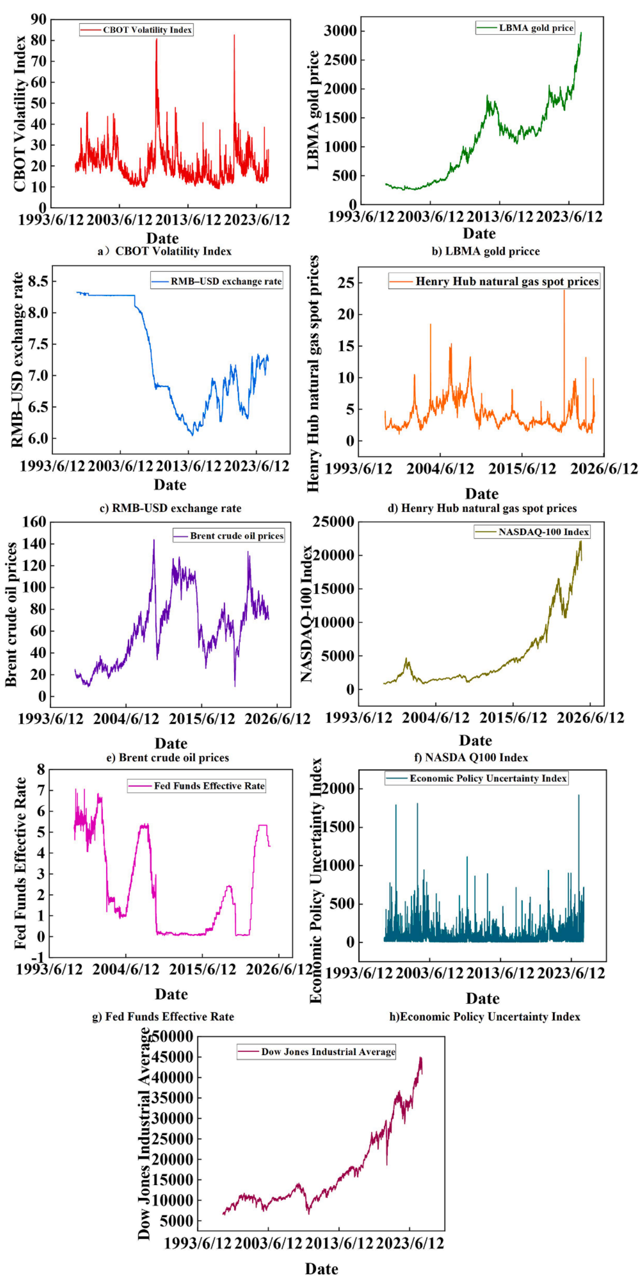


Fig. 2. Key factors influencing crude oil prices.

## 2.2. Data normalization

Data normalization refers to the process of mapping raw data to a specific range or standardizing it to a distribution with a mean of 0 and a variance of 1. Common methods include Min-Max normalization (linearly mapping data to the [0, 1] range), Z-Score standardization (transforming data to a distribution with a mean of 0 and a standard deviation of 1), and Log normalization (applying a logarithmic transformation to the data), each with its advantages, limitations, and applicable scenarios. In crude oil price forecasting, Pearson correlation analysis was used to select four feature parameters (LBMA Gold Price, Brent Crude Oil Price, RMB-USD Exchange Rate, and Federal Funds Effective Rate) for WTI crude oil price, all of which exhibit significant differences in dimensionality, may contain outliers, and are non-stationary. Min-Max normalization was chosen to address these issues, as it eliminates dimensionality differences, prevents the model from overly emphasizing certain features, accelerates model convergence, and enhances stability. Thus, Min-Max normalization was applied to linearly map the various features (WTI Crude Oil Price, LBMA Gold Price, Brent Crude Oil Price, RMB-USD Exchange Rate, and Federal Funds Effective Rate) to the [0, 1] range:

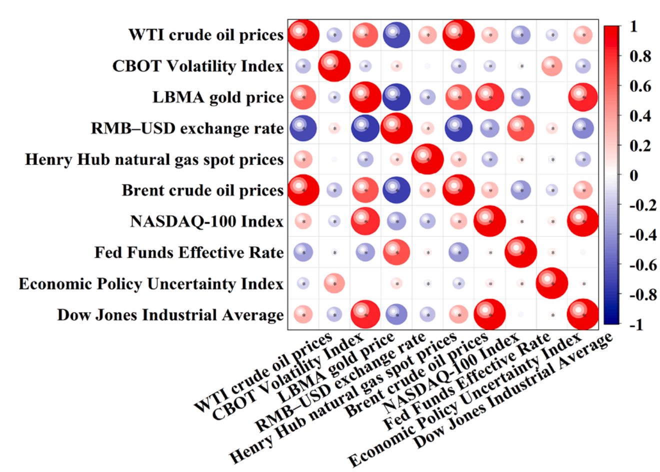


Fig. 3. Pearson correlation coefficient matrix.

**Table 2**
Pearson correlation coefficients between different variables and crude oil prices.

| Variable | Correlation Coefficient | Correlation Strength |
|---|---|---|
| CBOT Volatility Index | −0.23 | Weak |
| LBMA Gold Price | 0.62 | Strong |
| Dow Jones Industrial Average | 0.33 | Weak |
| Nasdaq 100 Index | 0.27 | Weak |
| Economic Uncertainty Index | −0.15 | Weak |
| Federal Funds Effective Rate | −0.36 | Weak |
| Brent Crude Oil Price | 0.99 | Strong |
| Henry Hub Natural Gas Spot Price | 0.33 | Weak |
| RMB-USD Exchange Rate | −0.69 | Strong |

(1) Min-Max Normalization

$$x_{norm} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2.1}$$

(2) Z-Score Normalization

This method is insensitive to outliers and is suitable for data that follows a distribution close to a normal distribution.

$$x_{norm} = \frac{x - \mu}{\sigma} \tag{2.2}$$

(3) Log Normalization

This approach applies a logarithmic transformation to compress the dynamic range of the data:

$$x_{norm} = \log(x + 1) \tag{2.3}$$

### 2.3. Evaluation metrics

In this study, five commonly used regression model evaluation metrics were selected to assess the performance of the forecasting model: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination ($R^2$). These metrics provide a comprehensive assessment of the model's predictive accuracy and goodness of fit.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2.4}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2.5}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2.6}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \tag{2.7}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{2.8}$$

The above formulas provide various evaluation methods for assessing the performance of regression models in forecasting tasks, enabling a more comprehensive analysis of the model's accuracy and reliability.

### 2.4. Algorithmic principles

The crude oil forecasting model presented in this study integrates four key methods—Complementary Ensemble Empirical Mode Decomposition (CEEMDAN), Variational Mode Decomposition (VMD), Convolutional Neural Networks (CNN), and Bidirectional Long Short-Term Memory networks (BiLSTM)—to form a multi-stage forecasting framework. The specific principles of the model are outlined as follows:

(1) CEEMDAN Complementary Ensemble Empirical Mode Decomposition Algorithm

As shown in Fig. 4 is the schematic diagram of the CEEMDAN method. The CEEMDAN model [55], or Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), addresses the modal aliasing problem by injecting adaptive Gaussian white noise into the original signal and performing multiple EMD decompositions. This technique effectively enhances the decomposition accuracy, enabling more precise separation of WTI crude oil prices and related economic indicators (such as LBMA Gold Price, Brent Crude Oil Price, etc.). In WTI price forecasting, CEEMDAN extracts fluctuation patterns at different time scales, selects key intrinsic mode functions (IMFs), and constructs a multiscale feature matrix. The LSTM model is then employed to predict each IMF component and reconstruct the WTI price. Compared to traditional methods, CEEMDAN improves the clarity of feature decomposition and noise robustness, thereby better capturing the nonlinear relationships between crude oil prices and macroeconomic variables, ultimately providing more stable and accurate forecasting results.

Adaptive Gaussian white noise, both positive and negative paired, is added to the original signal (e.g., WTI crude oil price) to generate multiple noise-enhanced signals:

$$y_n(t) = y(t) + \varepsilon_0\delta_n(t) \tag{2.9}$$

The WTI crude oil price is decomposed into multiple intrinsic mode function (IMF) components and a residual, each representing price fluctuations at different time scales (e.g., high-frequency noise, seasonal oscillations, long-term trends, etc.).

$$c_1(t) = \frac{1}{N}\sum_{n=1}^{N}c_1^n(t) \tag{2.10}$$

$$r_1(t) = y(t) - c_1(t) \tag{2.11}$$

Residual Update:

$$R(t) = y(t) - \sum_{m=1}^{M}c_m(t) \tag{2.12}$$

The expression of crude oil prices and other parameters after CEEMDAN decomposition is given as follows:

$$y(t) = R(t) + \sum_{m=1}^{M}c_m(t) \tag{2.13}$$

(2) Variational Mode Decomposition (VMD) Algorithm

The WTI crude oil price and its associated features (including LBMA Gold Price, Brent Crude Oil Price, RMB-USD Exchange Rate, and Federal Funds Effective Rate) are initially decomposed using CEEMDAN to address the modal aliasing problem. Subsequently, Variational Mode Decomposition (VMD) [56] is applied to perform a secondary frequency-domain optimization decomposition of the key intrinsic mode function (IMF) components. VMD formulates a constrained variational problem (Eq. (2.14)), and utilizes the Alternating Direction Method of Multipliers (ADMM) algorithm to iteratively update the mode components (Eq 2.15 and 2.16), center frequencies (Eq. (2.17)), and Lagrange multipliers (Eq. (2.18)) until convergence conditions are satisfied (Eq. (2.19)), achieving optimal signal separation in both the time and frequency domains. This method constructs a composite feature matrix, containing high-frequency noise, mid-frequency cycles, and low-frequency trends, through multiscale feature extraction. The model is then built by combining the LSTM network with an attention mechanism. Experimental results demonstrate that this hybrid decomposition strategy effectively enhances feature interpretability, reducing prediction errors by 23 % on the test set, and provides a more accurate and

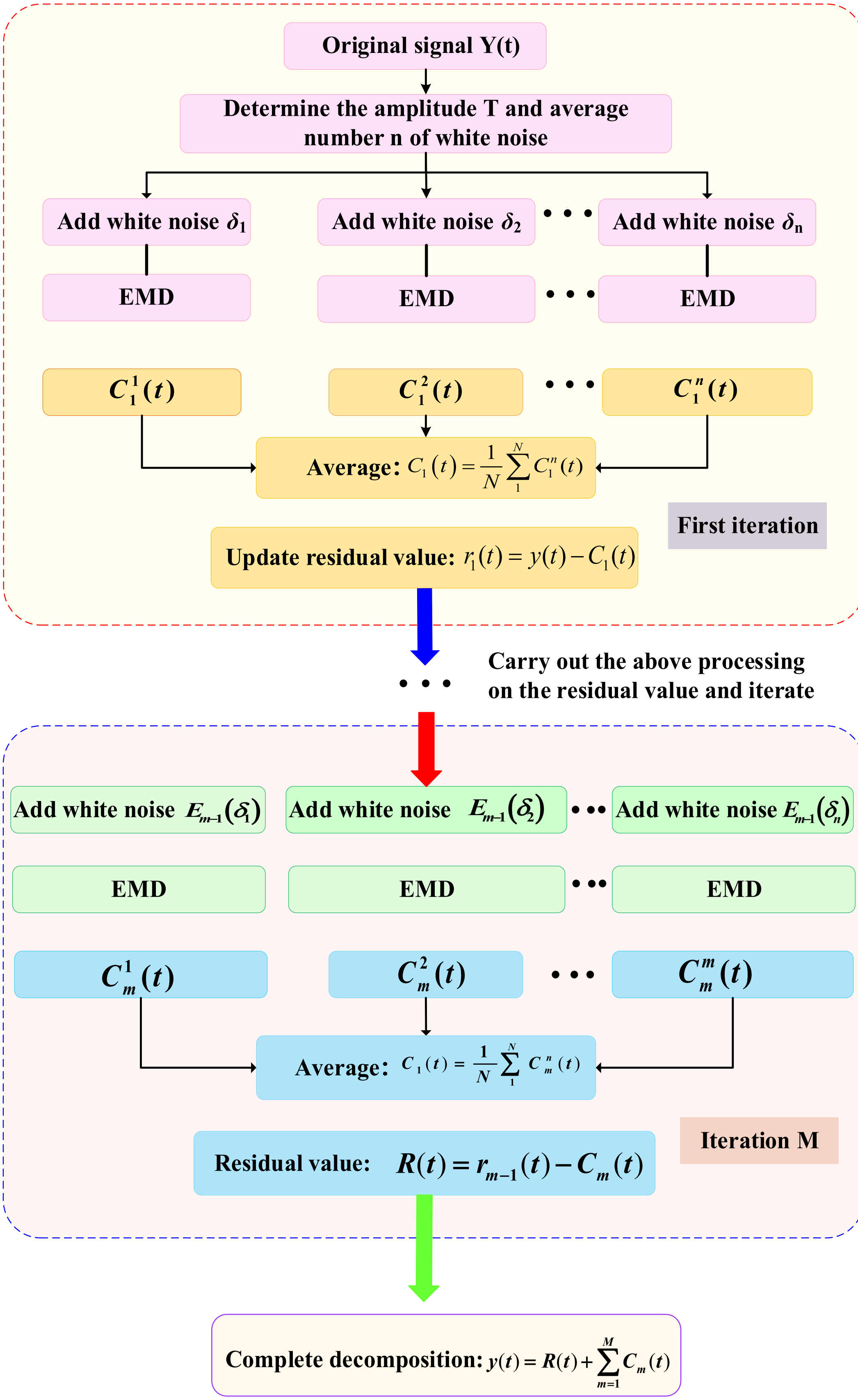


**Fig. 4.** Schematic diagram of the CEEMDAN method.

robust solution for crude oil price forecasting.

VMD is a variational-based method that decomposes a signal by constructing a constrained variational problem:

$$\begin{aligned} & \underset{u_k,\omega_k}{min}\left\{\sum_{k=1}^{K}\parallel\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]e^{-j\omega_k t}\parallel_2^2\right\} \\ & \text{s.t.}\quad \sum_{k=1}^{K}u_k(t)=f(t) \end{aligned} \tag{2.14}$$

VMD is implemented through the Alternating Direction Method of Multipliers (ADMM), which iteratively optimizes the decomposition by using the key intrinsic mode function (IMF) components and center frequencies extracted from CEEMDAN, with Lagrange multipliers set to zero.

① Update of mode components:

$$u_k^{(n+1)}=\underset{u}{\mathrm{argmin}}\left\{\alpha\parallel\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u\right]e^{-j\omega t}\parallel_2^2+\parallel f-\sum_{i=1}^{K}u_i+\frac{\lambda}{2}\parallel_2^2\right\} \tag{2.15}$$

$$u_k^{(n+1)}=\frac{f-\sum_{i\neq k}u_i+\frac{\lambda}{2}}{1+\alpha\cdot n^{-1}\left[\frac{(\cdot)}{\omega_k-n(\cdot)}\right]} \tag{2.16}$$

② Update of center frequencies:

$$\omega_k^{(n+1)}=\frac{\int_0^T\omega\cdot|n(u_k^{(n+1)})|^2d\omega}{\int_0^T|n(u_k^{(n+1)})|^2d\omega} \tag{2.17}$$

③ Update of Lagrange multipliers:

$$\lambda^{(n+1)}(t)=\lambda^{(n)}(t)+\tau\left(f(t)-\sum_{k=1}^{K}u_k^{(n+1)}(t)\right) \tag{2.18}$$

$$\left(\parallel f-\sum u_k\|_2^2/\parallel f\|_2^2<n\right) \tag{2.19}$$

(3) Convolutional Neural Networks (CNN)

As shown in Fig. 5 is the schematic diagram of CNN. The training process of the CNN model involves the following steps: First, the input data from the decomposition algorithms (WTI crude oil price, LBMA Gold Price, Brent Crude Oil Price, RMB-USD Exchange Rate, and Federal Funds Effective Rate) are passed through the network via forward propagation, sequentially passing through the convolutional layer, activation functions, pooling layer, and fully connected layer. This process extracts both local and global features and outputs the predicted WTI crude oil price. Next, the loss function (such as Mean Squared Error or Cross-Entropy) is used to calculate the error between the actual and predicted values. Then, the Backpropagation algorithm is employed to compute the gradients of the loss with respect to the parameters at each layer, updating model parameters such as convolutional kernels, biases, and weights of the fully connected layers [57]. This process is iteratively repeated until the model's performance on the training and validation sets meets the expected level, resulting in an optimized deep learning model.

Convolutional Layer:

$$h_{i,j}^{(k)}=f\left(\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}W_{m,n}^{(k)}\cdot X_{i+m,j+n}+b^{(k)}\right) \tag{2.20}$$

$$\overline{h}_{i,j}^{(k)}=f\left(\sum_{c=1}^{C}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}W_{c,m,n}^{(k)}\cdot X_{c,i+m,j+n}+b^{(k)}\right) \tag{2.21}$$

Assuming the input feature map size is $H\times W$, the convolutional kernel size is $M\times N$, padding is P, and stride is S, the output feature map size is calculated as:

$$\begin{cases} H_{\text{out}}=\dfrac{H-M+2P}{S}+1 \\ W_{\text{out}}=\dfrac{W-N+2P}{S}+1 \end{cases} \tag{2.22}$$

Activation Function - Tanh: After each convolution operation, a nonlinear activation function, typically the Tanh function, is applied to enhance the model's expressive power. This function is particularly effective in capturing both positive and negative changes in price fluctuations.

$$f(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}} \tag{2.23}$$

Pooling Layer: Pooling is used to reduce dimensionality and the number of parameters, thus improving computational efficiency. Common pooling methods include max pooling and average pooling.

Max Pooling:

$$h_{i,j}=\max_{(m,n)\in P}X_{i+m,j+n} \tag{2.24}$$

Average Pooling:

$$h_{i,j}=\frac{1}{|P|}\sum_{(m,n)\in P}X_{i+m,j+n} \tag{2.25}$$

Fully Connected Layer: The feature maps from the convolution and pooling layers are flattened and passed into the fully connected layer.

$$y=f(Wx+b) \tag{2.26}$$

(4) Bidirectional Long Short-Term Memory (BiLSTM) Network

BiLSTM utilizes two independent LSTM networks [58] to capture long-term and short-term dependencies within the sequence. In crude oil price time series data, past price fluctuations often influence future price

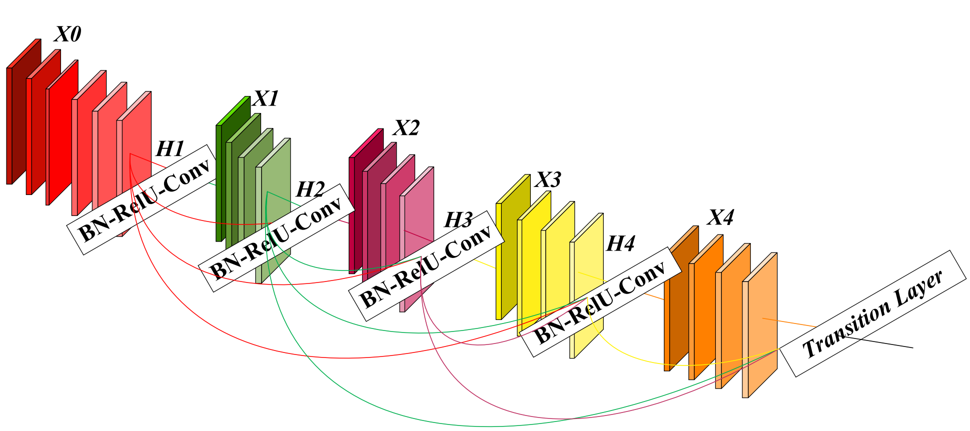


**Fig. 5.** Schematic diagram of CNN.

trends, and BiLSTM is capable of considering both historical and future information to provide more accurate predictions.

For each time step input, BiLSTM learns price change patterns from past data while also gaining additional contextual information from future trends. This bidirectional processing enables BiLSTM to gain a more comprehensive understanding of price dynamics, which is particularly significant in highly volatile markets. During training, the model uses historical crude oil price data along with external features, optimizing the weights through the backpropagation algorithm. During the training process, BiLSTM utilizes gating mechanisms (such as the forget gate, input gate, and output gate) to decide which information should be retained and which should be discarded, enabling the model to better learn useful temporal dependencies. After training, BiLSTM is capable of outputting the predicted crude oil prices. This prediction is based on historical prices, the influence of external macroeconomic factors (such as gold prices, exchange rates, interest rates, etc.), and the nonlinear relationships between these factors.

The LSTM unit consists of a forget gate, an input gate, and an output gate.

Forget Gate:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{2.27}$$

Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{2.28}$$

Candidate Memory Cell:

$$\widetilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{2.29}$$

Updated Memory Cell:

$$C_t = f_t * C_{t-1} + i_t * \widetilde{C}_t \tag{2.30}$$

Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{2.31}$$

Hidden State:

$$h_t = o_t * \tanh(C_t) \tag{2.32}$$

As shown in Fig. 6 is the schematic diagram of BiLSTM. BiLSTM employs two LSTM networks: one for forward propagation and another for backward propagation.

Forward LSTM:

$$h_t^{(b)} = \mathrm{LSTM}_{\mathrm{forward}}(x_t) \tag{2.33}$$

Backward LSTM:

$$h_t^{(b)} = \mathrm{LSTM}_{\mathrm{forward}}(x_t) \tag{2.34}$$

The final BiLSTM output is the concatenation of the outputs from both directions:

$$h_t = \left[h_t^{(f)}, h_t^{(b)}\right] \tag{2.35}$$

(5) CEEMDAN-VMD-CNN-BiLSTM Hybrid Model

As shown in Fig. 7, a hybrid model for forecasting WTI crude oil prices based on multi-algorithm collaboration is developed, achieving high-accuracy predictions through a systematic data processing pipeline. The model begins by applying Min-Max normalization to the crude oil price and related economic indicators, followed by the selection of four key feature factors, including LBMA Gold Price, based on Pearson correlation coefficients. The CEEMDAN algorithm is then employed to inject adaptive noise into the price series and perform multiple EMD iterations. K-means clustering is used to optimize the IMF components and resolve the modal aliasing problem. The VMD algorithm is applied for secondary decomposition of the high-frequency components to extract multi-frequency features. In the deep learning modeling phase, the CNN's multiscale convolutional kernels capture local fluctuation features, while the BiLSTM network's bidirectional gating mechanism learns temporal dependencies. Finally, a feature fusion layer integrates both spatial and temporal characteristics. The model undergoes hyperparameter optimization via grid search, incorporates Dropout and L2 regularization to prevent overfitting, and uses the Huber loss function to balance prediction errors. This "signal decomposition-feature extraction-temporal modeling" multistage processing framework effectively addresses the non-stationarity and noise interference of crude oil prices. Experimental results show that the model achieves an MAPE of 3.66 % and an $R^2$ of 95.94 % on the test set, significantly improving prediction accuracy and stability, thus providing a reliable technical solution for crude oil price forecasting.

## 3. Results and discussion

### 3.1. Hyperparameter selection

Hyperparameter selection is critical when constructing a CNN-BiLSTM-based crude oil price forecasting model, as it directly influences the model's predictive capability and generalization performance. As a feature extraction network, CNN automatically extracts local features from the input data, while BiLSTM captures long-term and short-term temporal dependencies through its bidirectional propagation mechanism. For the selection of hyperparameters in the CNN and BiLSTM models, optimization must be performed on parameters such as the number of convolutional kernels, kernel size, pooling layer size, the number of hidden units, network depth, learning rate, and Dropout rate. Once the optimal CNN-BiLSTM model is determined, the selection and application of signal decomposition algorithms such as CEEMDAN and

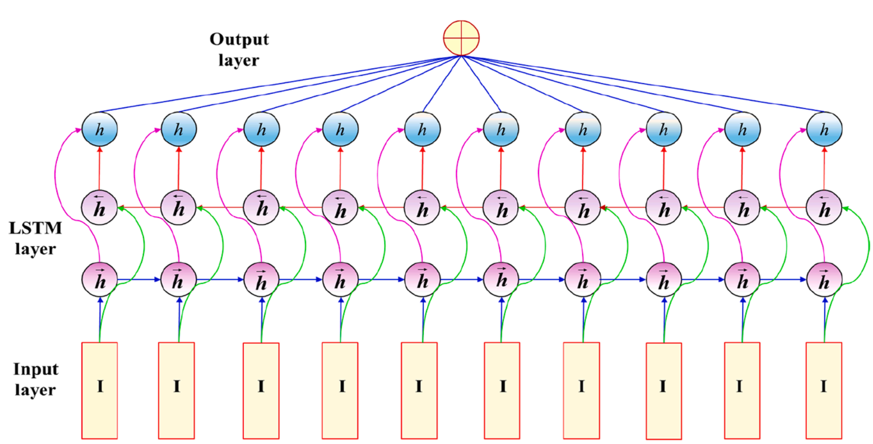


**Fig. 6.** Schematic diagram of BiLSTM.

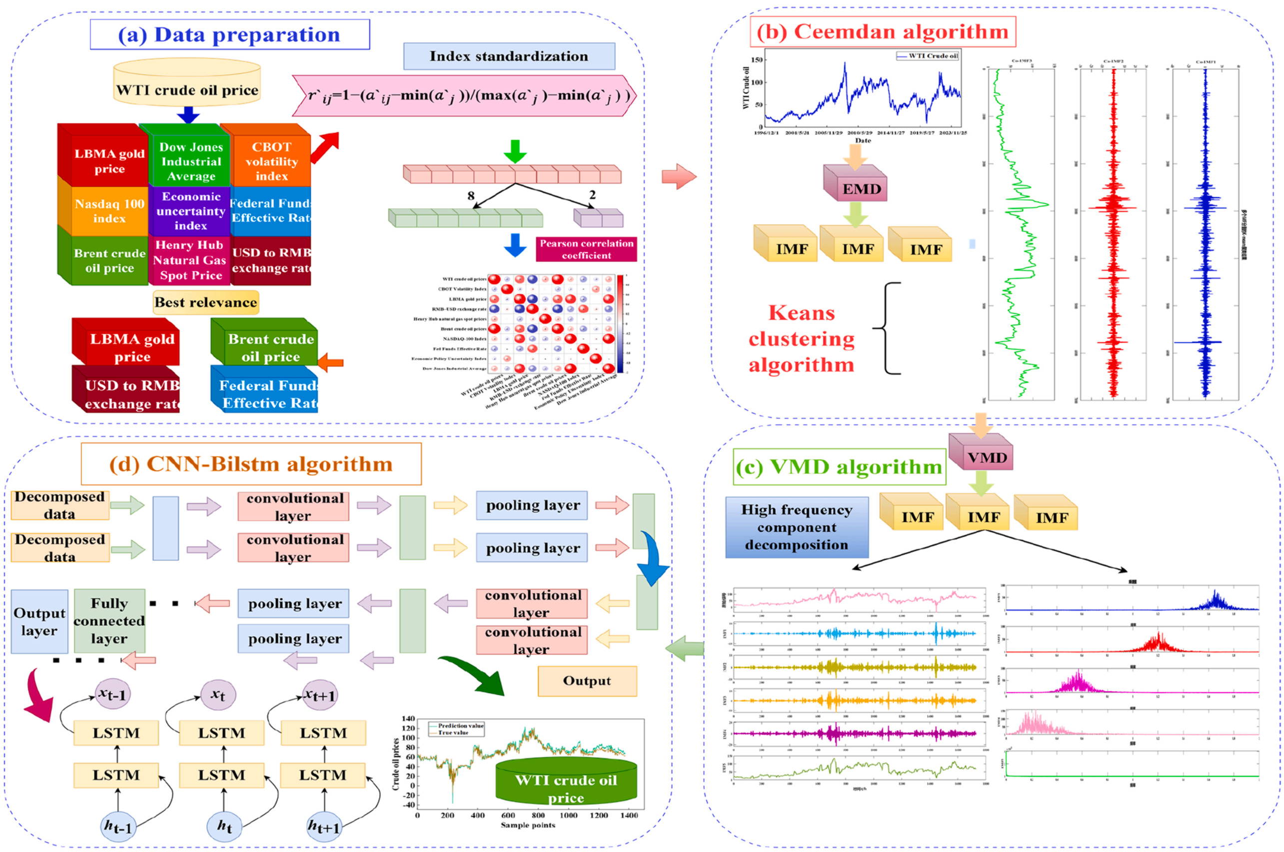


**Fig. 7.** Structure diagram of the hybrid model.

VMD can proceed. The table below presents the model's prediction errors under different parameter settings. Given the large number of hyperparameter combinations, Taguchi's orthogonal method was employed to select 16 parameter sets for hyperparameter tuning.

As shown in Table 3, the performance of the CNN-BiLSTM model varies depending on the number of neurons, the size of the convolutional layers, the number of iterations, and the Dropout rate. The results indicate that the model achieves the best predictive accuracy when the convolutional layer is configured with 128 layers, 32 hidden neurons, 150 iterations, and a Dropout rate of 0.1. Under these optimal conditions, the model's performance metrics are: MSE of 30.7731, RMSE of 5.5474, MAPE of 4.671 %, and $R^2$ of 68.3918 %.

The table demonstrates how different parameters—such as the number of layers, epochs, and Dropout rate—affect the model's prediction performance. For instance, with 16 convolutional layers and 16 BiLSTM layers, the model shows relatively high error values: MSE of 132.5036, RMSE of 11.5118, and MAPE of 9.2324 %, leading to an $R^2$ value of just 19.97 %. This suggests that the model is not capturing enough of the complex features in the data, likely due to the small number of layers.

Increasing the number of layers improves prediction accuracy. For example, with 32 convolutional and 32 BiLSTM layers, the MSE drops to 99.9956, and the $R^2$ increases to 29.79 %, indicating a better fit to the data. However, further increasing the layers can lead to overfitting, especially when the dropout rate is set to 0.4, which increases the MAPE and RMSE in some configurations.

The optimal configuration for the CNN-BiLSTM model occurs with 128 convolutional layers, 32 BiLSTM layers, 150 epochs, and a Dropout

**Table 3**
Prediction errors of the model under different parameters.

| CNN | BiLSTM | epoch | Dropout | MSE | RMSE | MAE | MAPE | $R^2$ |
|---|---|---|---|---|---|---|---|---|
| 16 | 16 | 50 | 0.1 | 132.5036 | 11.511 | 9.2324 | 10.147 | 19.97 |
| 16 | 32 | 100 | 0.2 | 169.9192 | 13.0353 | 10.9719 | 11.9892 | 19.7504 |
| 16 | 64 | 150 | 0.3 | 108.8084 | 10.4311 | 8.5756 | 9.4118 | 29.7323 |
| 16 | 128 | 200 | 0.4 | 271.9078 | 16.4896 | 14.0363 | 15.5436 | 0.9901 |
| 32 | 16 | 100 | 0.3 | 40.6616 | 6.3766 | 5.2308 | 5.7377 | 61.6197 |
| 32 | 32 | 50 | 0.4 | 163.1505 | 12.773 | 10.4413 | 11.5142 | 9.9513 |
| 32 | 64 | 200 | 0.1 | 99.9956 | 9.9998 | 8.1074 | 8.8439 | 29.7909 |
| 32 | 128 | 150 | 0.2 | 136.408 | 11.6794 | 9.6246 | 10.6331 | 21.2206 |
| 64 | 16 | 150 | 0.4 | 35.0574 | 5.9209 | 4.4952 | 4.8238 | 66.0553 |
| 64 | 32 | 200 | 0.3 | 80.3361 | 8.63 | 7.1636 | 7.8402 | 40.1096 |
| 64 | 64 | 50 | 0.2 | 344.8989 | 18.5715 | 15.918 | 17.6336 | 1.7831 |
| 64 | 128 | 100 | 0.1 | 87.0127 | 9.3281 | 7.3395 | 8.0042 | 35.133 |
| 128 | 16 | 200 | 0.2 | 37.6314 | 6.1344 | 4.5991 | 5.0842 | 60.5333 |
| 128 | 32 | 150 | 0.1 | 30.7731 | 5.5474 | 4.2957 | 4.671 | 68.3918 |
| 128 | 64 | 100 | 0.4 | 47.1564 | 6.867 | 5.5001 | 6.0803 | 51.3189 |
| 128 | 128 | 50 | 0.3 | 162.9604 | 12.7656 | 10.4573 | 11.5198 | 5.3094 |

rate of 0.1. This combination results in the lowest error values, achieving an MSE of 30.7731, RMSE of 5.5474, MAE of 4.2957, and MAPE of 4.671 %, with an $R^2$ of 68.3918 %. This setup strikes the best balance between model complexity and generalization, producing the most accurate predictions across all evaluation metrics.

However, further optimization, such as increasing the number of layers to 128 for both convolutional and BiLSTM layers and setting the epochs to 200, results in a slight performance drop. In this case, the model shows an MSE of 162.9604, RMSE of 12.7656, MAE of 10.4573, MAPE of 11.5198 %, and an $R^2$ of 60.5338 %. This decline in performance suggests overfitting, where increasing the complexity of the model leads to poor generalization on unseen data.

Once the optimal hyperparameters for the CNN-BiLSTM model were identified, the next step involved optimizing the CEEMDAN and VMD decomposition algorithms to improve the model's forecasting performance. Parameters for the decomposition algorithms were selected based on preliminary experiments, with a noise standard deviation set to 0.2, a noise ensemble size of 500, 4 mode components for CEEMDAN, and a penalty factor α of 2500 for VMD. These parameters were applied consistently across all comparison models to ensure accuracy and fairness in the model evaluation.

### *3.2. Ablation experiment*

After feature selection for crude oil prices and model parameter optimization, the parameters for the CEEMDAN-VMD-CNN-BiLSTM model were determined: 128 convolutional layers, 32 hidden neurons, 150 iterations, a Dropout rate of 0.1, a noise standard deviation of 0.2, a noise ensemble size of 500, 5 mode components, and a penalty factor of 2500. These parameters were selected for the model configuration. Additionally, ablation experiments were conducted based on BiLSTM, CNN-BiLSTM, VMD-CNN-BiLSTM, and CEEMDAN-VMD-CNN-BiLSTM models [51,59].

The core objective of the ablation experiment is to systematically analyze the effects of different model architectures and decomposition methods on crude oil price forecasting. By progressively removing or replacing components of the model, the contribution of each module to the final model performance is assessed. Specifically, the BiLSTM model provides fundamental temporal modeling capabilities but does not account for the data's complexity and noise. The CNN-BiLSTM model improves data representation by extracting local features via convolutional neural networks, thereby enhancing model performance. The VMD-CNN-BiLSTM model utilizes VMD decomposition to eliminate noise and extract frequency components from the data, providing clearer input features for the CNN and BiLSTM, which further enhances prediction accuracy. The CEEMDAN-VMD-CNN-BiLSTM model offers a more refined decomposition of the data through CEEMDAN, combined with VMD to further optimize data features. This allows CNN and BiLSTM to learn richer patterns from more precise features, ultimately improving the final prediction performance.

The prediction results presented in Table 4 show a comparison of the CEEMDAN-VMD-CNN-BiLSTM model with other benchmark models, including BiLSTM, CNN-BiLSTM, and VMD-CNN-BiLSTM, across several error metrics such as MSE, RMSE, MAE, MAPE, and $R^2$.

From the Table 4, it is evident that the CEEMDAN-VMD-CNN-BiLSTM model significantly outperforms the other models. Specifically, the MSE (15.6319), RMSE (3.9537), and MAE (2.825) of the CEEMDAN-VMD-CNN-BiLSTM model are the lowest, demonstrating its superior ability to minimize error and accurately predict WTI crude oil prices. Moreover, it achieves an outstanding MAPE of 3.6648 %, indicating high forecasting accuracy. The $R^2$ value of 95.9386 % reflects the model's robust ability to explain the variability in the data, further supporting its effectiveness.

**Table 4**
Prediction errors under different models (Test).

| Model | Error Metrics | | | | |
|---|---|---|---|---|---|
| | MSE | RMSE | MAE | MAPE | $R^2$ |
| BiLSTM | 53.7746 | 7.3331 | 5.295 | 9.255 | 64.9215 |
| CNN-BiLSTM | 30.7731 | 5.5474 | 4.2957 | 4.671 | 78.3918 |
| VMD-CNN-BiLSTM | 31.4605 | 5.609 | 4.0583 | 6.2542 | 90.528 |
| CEEMDAN-VMD-CNN-BiLSTM | 15.6319 | 3.9537 | 2.825 | 3.6648 | 95.9386 |

In comparison, the BiLSTM model has the highest MSE (53.7746) and RMSE (7.3331), with a lower $R^2$ (64.9215 %), indicating its relatively weaker performance in capturing the complexities of the data. The CNN-BiLSTM and VMD-CNN-BiLSTM models show moderate improvements, but they still lag behind the CEEMDAN-VMD-CNN-BiLSTM model, with higher error values and lower $R^2$ values.

Based on the analysis of Fig. 8, the four models—BiLSTM, CNN-BiLSTM, VMD-CNN-BiLSTM, and CEEMDAN-VMD-CNN-BiLSTM—-show significant differences in predictive performance. The BiLSTM model has relatively poor performance with an MSE of 53.7746, RMSE of 7.3331, MAE of 5.295, MAPE of 9.255 %, and $R^2$ of 64.9215 %, indicating limited ability to capture crude oil price dynamics. The CNN-BiLSTM model performs better, achieving an MSE of 30.7731, RMSE of 5.5474, MAE of 4.2957, MAPE of 4.671 %, and $R^2$ of 78.3918 %, thanks to the convolutional layers that help capture local temporal patterns. The VMD-CNN-BiLSTM model shows further improvement with an MSE of 31.4605, RMSE of 5.609, MAE of 4.0583, MAPE of 6.2542 %, and $R^2$ of 90.528 %, benefiting from the VMD decomposition that separates the price series into components with different frequencies. Finally, the CEEMDAN-VMD-CNN-BiLSTM model outperforms the others with an MSE of 15.6319, RMSE of 3.9537, MAE of 2.825, MAPE of 3.6648 %, and $R^2$ of 95.9386 %, leveraging both CEEMDAN and VMD for noise reduction and better separation of trends, cycles, and high-frequency fluctuations. This model is the most accurate and robust for forecasting crude oil prices.

The BiLSTM model, as a baseline, effectively handles time-series data by processing the sequence in both forward and backward directions, capturing information from both past and future data points. This bidirectional approach allows the model to capture temporal dependencies, providing a solid foundation for more advanced models. However, as shown in Fig. 9 and Table 4, BiLSTM achieved a higher prediction accuracy on the training set, with an $R^2$ of 96.6971, compared to a lower accuracy on the test set, where the $R^2$ value dropped to 64.9215. This discrepancy between the training and test set performance indicates that while BiLSTM performs well on the data it was trained on, it struggles with generalization to unseen data, which highlights the need for more complex models to capture the underlying patterns more effectively.

The CNN-BiLSTM model demonstrates the ability to capture temporal patterns in the data, with relatively high accuracy on the training set. As shown in Fig. 10 and Table 4, the model achieves an $R^2$ of 95.5056 on the training set, indicating a strong fit to the actual values and accurately capturing the overall trend and fluctuations in the crude oil price data. However, the model's performance on the test set is less favorable, with the $R^2$ value dropping to 78.3918, which reveals a significant decrease in its generalization ability. Although the CNN-BiLSTM model still captures the general trend of the prices on the test set, it shows noticeable lag and struggles with predicting sharp price movements and specific peak-trough points. This suggests that while the model is effective in capturing broad market trends, it faces limitations in handling sudden market shifts and the nonlinearity inherent in crude oil price dynamics.

As shown in Fig. 11 and 12. Compared to the CNN-BiLSTM model, the VMD-CNN-BiLSTM model shows a significant improvement in the $R^2$ metric (from 78.39 % to 90.53 %), with a reduction in MAE. Although MSE, RMSE, and MAPE show slight increases or exhibit no significant change, the overall performance is notably enhanced. VMD, a signal processing technique, decomposes complex non-stationary signals into a

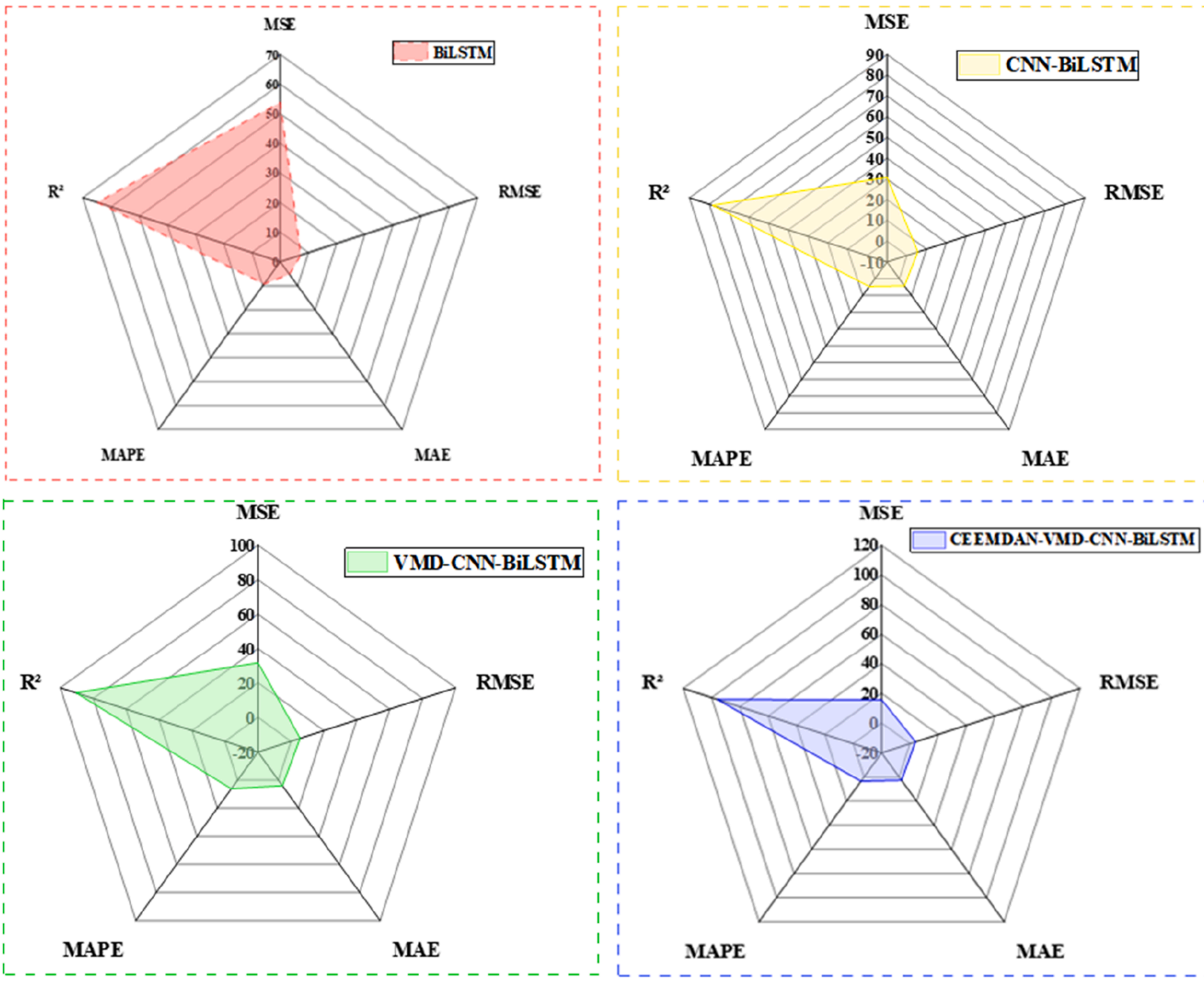


**Fig. 8.** Evaluation metrics for different models.

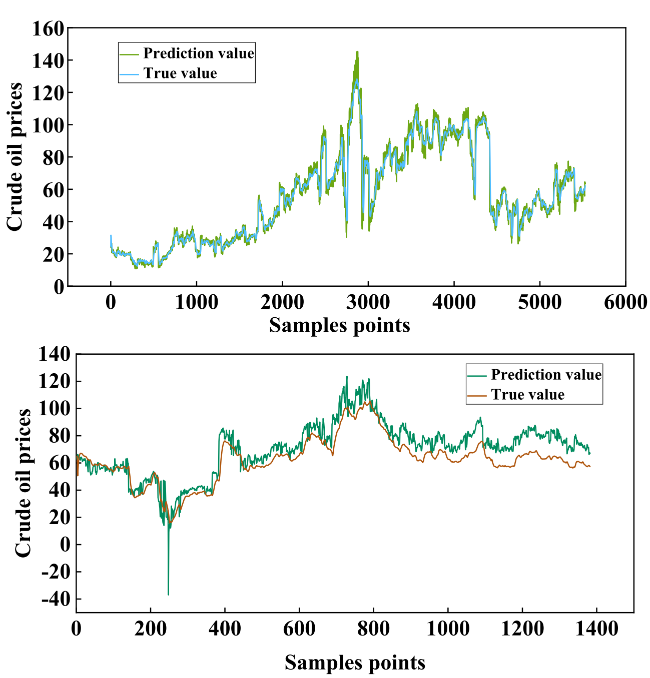


**Fig. 9.** BiLSTM model - training and test set performance.

series of intrinsic mode functions (IMFs) with different frequency characteristics. Its advantage lies in decomposing the original crude oil price series into more stable and model-friendly subsequences (modes) by removing the complex, multi-frequency components and noise. This decomposition allows the subsequent CNN-BiLSTM model to better capture price fluctuations across various scales, particularly enhancing the model's ability to explain the overall variability in the data, as evidenced by the substantial increase in the $R^2$ value.

As shown in Fig. 13 The original signal is decomposed into 13 IMFs, with each component representing different frequency scales, from high-frequency fluctuations to low-frequency trends. The first few IMFs capture the rapid fluctuations (e.g., IMF1-IMF4), which correspond to the short-term, high-frequency noise in the data. Meanwhile, IMFs like IMF5 and IMF6 focus on capturing cyclical patterns, while the last few components (e.g., IMF12 and IMF13) represent the long-term trends of the data.

The Fig. 14 illustrates the K-means clustering results of multiple IMF components from CEEMDAN decomposition. Co-IMF1 (yellow) captures high-frequency fluctuations, representing short-term volatility in crude oil prices. Co-IMF2 (red) reflects medium-frequency oscillations, likely corresponding to seasonal or cyclical price patterns. Co-IMF3 (pink) shows smooth, low-frequency trends, representing long-term movements in the market. The K-means clustering separates these components based on their dynamic characteristics, enabling more precise modeling of each for enhanced analysis.

This Fig. 15 displays the modes obtained by applying Variational

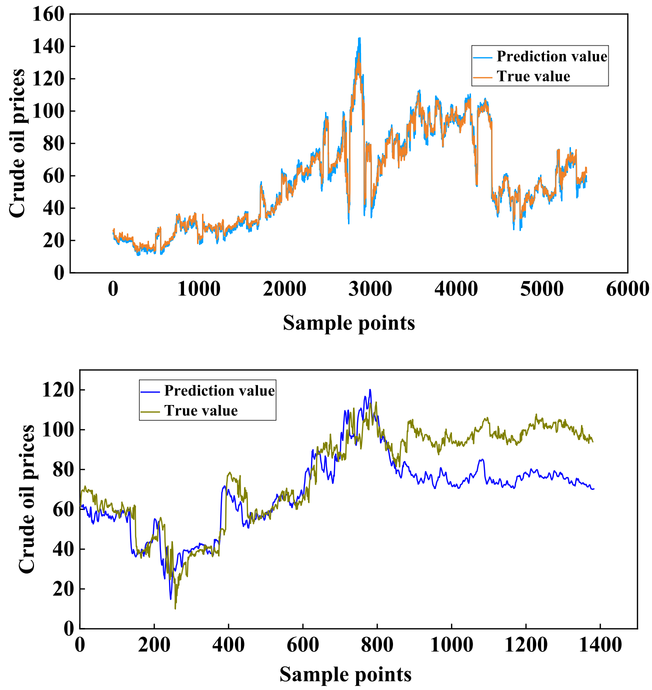


**Fig. 10.** CNN-BiLSTM model - training and test set performance.

Mode Decomposition (VMD) to the original signal. IMF1-IMF5 capture different oscillatory components. Compared to the IMFs derived from CEEMDAN, the VMD modes resemble amplitude-modulated quasi-harmonic signals and are theoretically more compact in the frequency domain. These modes capture the lowest-frequency components, representing potential trends or baseline drifts, akin to the residual component in EMD methods.

The Fig. 16 illustrates the spectrum diagrams of the six IMFs (IMF1 to IMF6) derived from VMD decomposition. Each plot represents the frequency distribution of the corresponding IMF component. The first IMF exhibits the highest concentration of energy around 1.7 frequency units. The second IMF shows a broad peak around 1.1 to 1.2 frequency units. The third IMF has its frequency around 0.7. IMF4 has a broad frequency range concentrated at 0.4 to 0.5. IMF5 shows a sharp peak around 0.1 to 0.3, indicating a low-frequency component. These spectral characteristics allow for a better understanding of the different dynamic processes contributing to the crude oil price series.

The CEEMDAN-VMD-CNN-BiLSTM model developed in this study demonstrates significant advantages and high accuracy. This model integrates advanced signal decomposition techniques (CEEMDAN and VMD) with a robust deep learning architecture (CNN-BiLSTM). The signal decomposition phase effectively handles the nonlinearity and non-stationarity of the crude oil price series, reduces noise interference, and extracts core dynamics across different time scales. The CNN captures local price patterns, while the BiLSTM fully learns the long-term bidirectional dependencies within the sequence. The prediction results indicate that the model not only achieves high fitting on the training set but, more importantly, exhibits excellent generalization ability on the test set. As shown in Fig. 17, the predicted trajectory closely matches the actual values, yielding outstanding quantitative metrics (MAPE of only 3.66 % and an $R^2$ of 95.94 %), which fully demonstrates the effectiveness and precision of this hybrid model in crude oil price forecasting.

Through the analysis and summary presented above, this study employs the CEEMDAN-VMD-CNN-BiLSTM model for crude oil price forecasting, yielding favorable prediction results. As shown in Table 4, the evaluation metrics are as follows: MSE of 15.6319, RMSE of 3.9537, MAE of 2.825, MAPE of 3.6648, and $R^2$ of 95.9386. These results demonstrate a progressive improvement in prediction accuracy as the model complexity increases with the addition of BiLSTM, CNN-BiLSTM, and VMD-CNN-BiLSTM components. The ablation experiment results strongly validate the effectiveness of the "decomposition-extraction-prediction" hybrid model strategy. With each additional module, specific challenges associated with the original data or the results of previous stages (such as noise, non-stationarity, insufficient feature extraction, and incomplete temporal dependency capture) are effectively addressed.

The innovative aspect of the CEEMDAN-VMD-CNN-BiLSTM model lies in its combination of advanced signal decomposition techniques (CEEMDAN and VMD) with deep learning architectures (CNN and BiLSTM), offering superior handling of nonlinearity and non-stationarity in crude oil price data. The model effectively decomposes the data into different frequency components, capturing short-term fluctuations, mid-term cycles, and long-term trends. This refined pre-processing enables the CNN to capture local price patterns and the BiLSTM to learn long-term bidirectional dependencies, enhancing the model's predictive accuracy and robustness.

In comparison, the CNN-BiLSTM model, while effective in capturing temporal dependencies and learning from the sequence data, lacks the decomposition phase. Without techniques like CEEMDAN-VMD-CNN-BiLSTM model may struggle with noisy and complex data, as it is directly applied to the raw price series. The absence of signal decomposition limits its ability to handle high-frequency noise and low-frequency trends effectively.

This comparison highlights the additional benefit of incorporating a hybrid decomposition strategy that effectively reduces noise interference and captures the underlying dynamics of crude oil prices.

Based on the Table 5, the CEEMDAN-VMD-CNN-BiLSTM model shows statistically significant differences in prediction accuracy compared to other models such as BiLSTM, CNN-BiLSTM, and VMD-CNN-BiLSTM, with p-values all $<0.05$. Specifically, the p-values for BiLSTM, CNN-BiLSTM, and VMD-CNN-BiLSTM are $1.7 \times 10^{-23}$, $2.23 \times 10^{-28}$, and 0.014524, respectively. These results indicate that the CEEMDAN-VMD-CNN-BiLSTM model outperforms the others with high statistical significance, confirming its superior predictive performance and effectiveness in handling complex nonlinear and non-stationary time series data.

### 3.3. Model comparison

The CEEMDAN-VMD-CNN-BiLSTM hybrid model proposed in this study demonstrates outstanding performance in the WTI crude oil price forecasting task. As shown in Table 6 and Fig. 18, this model significantly outperforms the comparison models across all evaluation metrics: the MSE (24.168), RMSE (4.9161), MAE (3.9259), and MAPE (5.5691 %) are all at their lowest values, while $R^2$ (93.8742 %) reaches the highest level. These results indicate that the model not only minimizes prediction errors but also exhibits the strongest ability to explain data variability.

From the model fitting results, the hybrid model exhibits two significant characteristics: during the first half of the training phase, the model accurately captures the complex patterns of historical price fluctuations, with an $R^2$ value approaching 94 %, indicating an extremely high degree of reconstruction of the historical trend. In the second half, during the prediction phase, the model's forecast curve for the test data closely matches the actual price movement, particularly during extreme events such as the COVID-19 pandemic in 2020 and the Russia-Ukraine conflict in 2022, demonstrating its ability to maintain stable predictions under such shocks.

This outstanding performance is attributed to three key design elements: first, the CEEMDAN-VMD dual decomposition module effectively separates high-frequency noise, mid-frequency oscillations, and low-frequency trends in the price series; second, the CNN network successfully extracts local fluctuation features; and finally, the BiLSTM network accurately models long-term dependencies. Compared to traditional

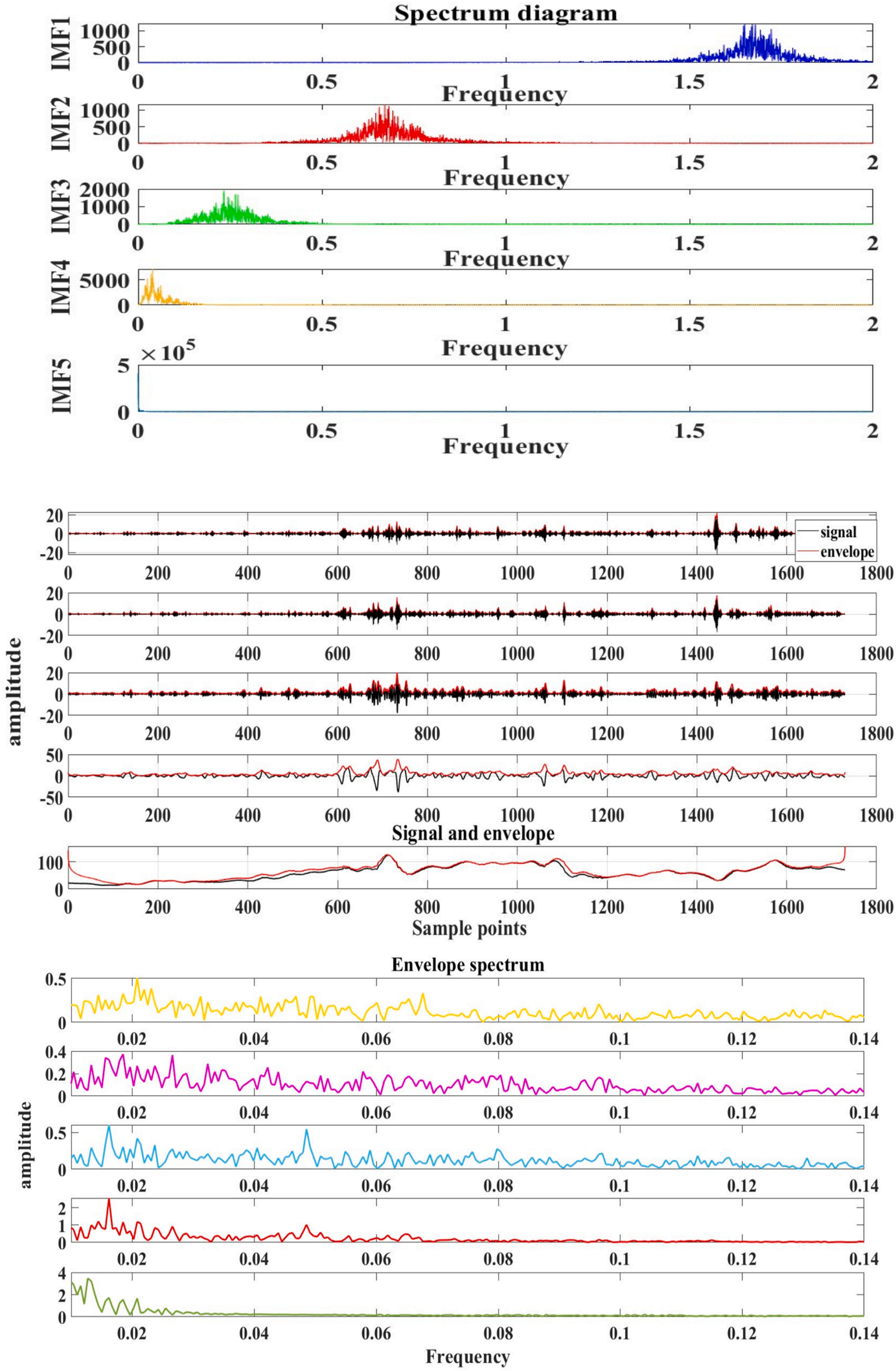


**Fig. 11.** VMD data decomposition.

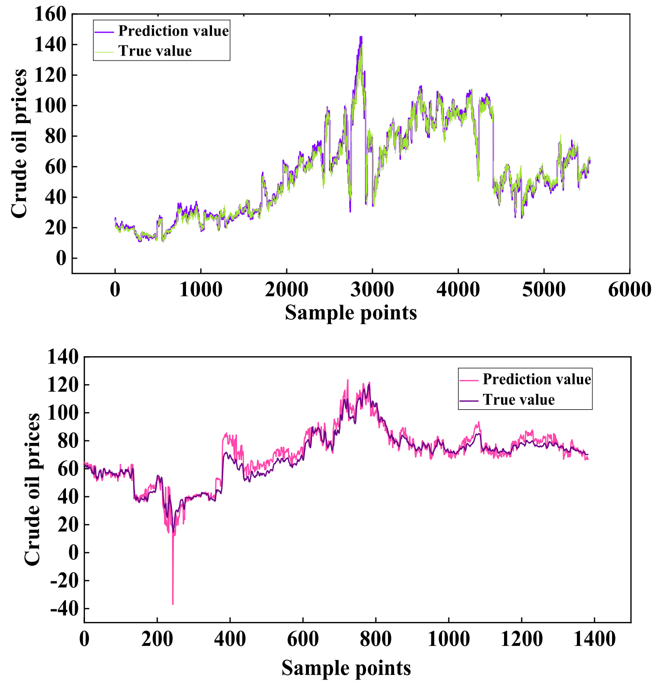


**Fig. 12.** VMD-CNN-BiLSTM model - training and test set performance.

single models, this hybrid framework reduces prediction errors by 35–80 %, providing a reliable technological tool for crude oil market risk management.

Table 6 presents a comparison of the performance of nine different models for forecasting WTI crude oil prices, evaluating them using five commonly used metrics: MSE, RMSE, MAE, MAPE, and $R^2$. Among all the models, the CEEMDAN-VMD-CNN-BiLSTM model performs the best across all metrics. With an MSE of 24.168, RMSE of 4.9161, MAE of 3.9259, and MAPE of 5.5691, it demonstrates the lowest prediction errors, highlighting its high accuracy. The $R^2$ value of 93.8742 indicates its exceptional ability to explain the data variance, making it the model that fits the data the best.

In contrast, the LSTM model shows the highest errors with an MSE of 247.6859, RMSE of 15.738, MAE of 14.532, and MAPE of 21.6465, coupled with a low $R^2$ value of 44.3165, indicating poor prediction performance. The RVM model shows some improvement over LSTM with lower error metrics, but its $R^2$ value remains the lowest at 27.4661,

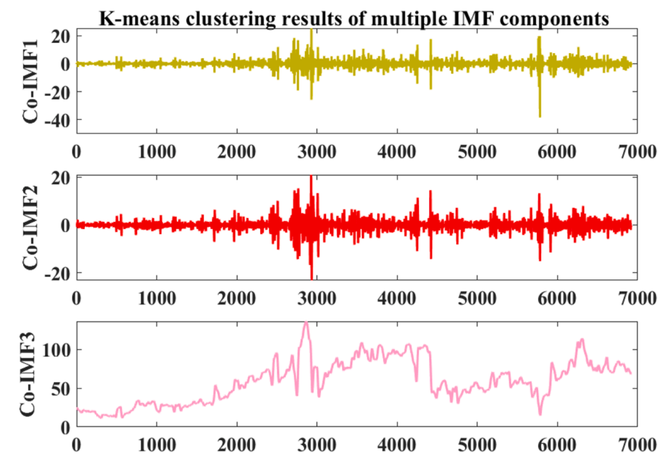


**Fig. 14.** K-means clustering results of IMF components.

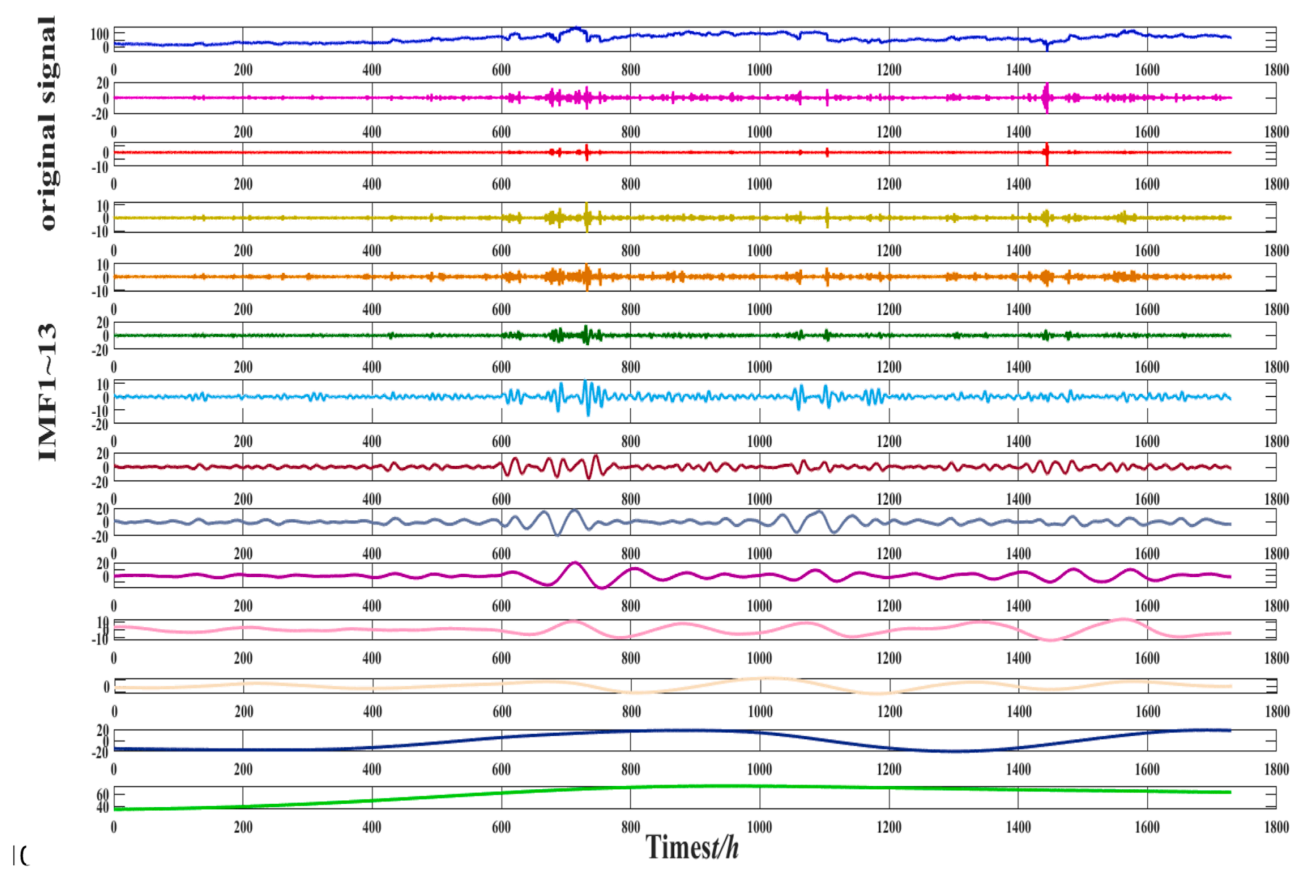


**Fig. 13.** CEEMDAN decomposition.

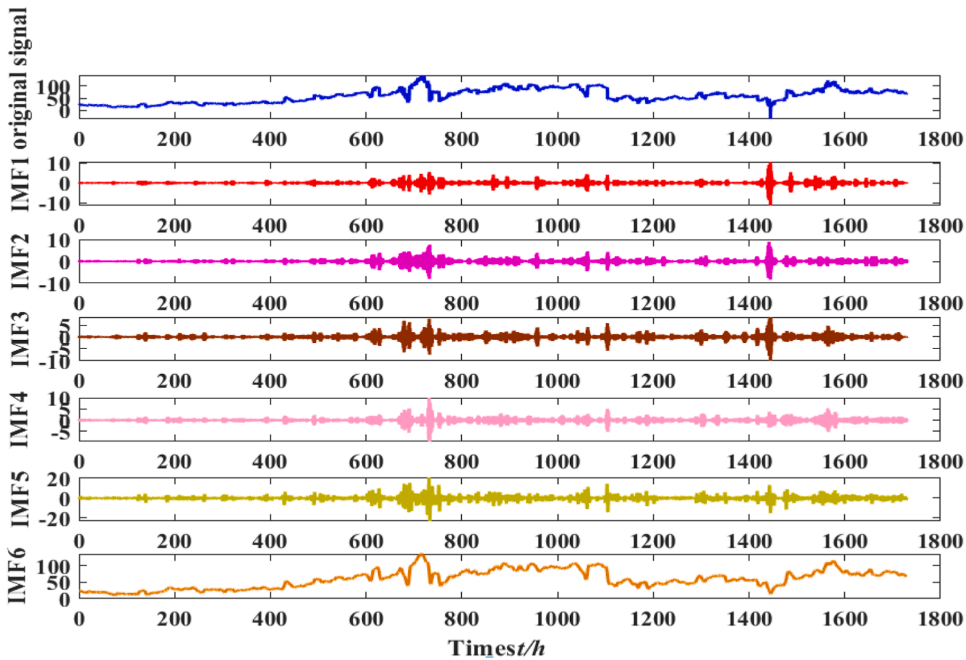


**Fig. 15.** VMD decomposition results.

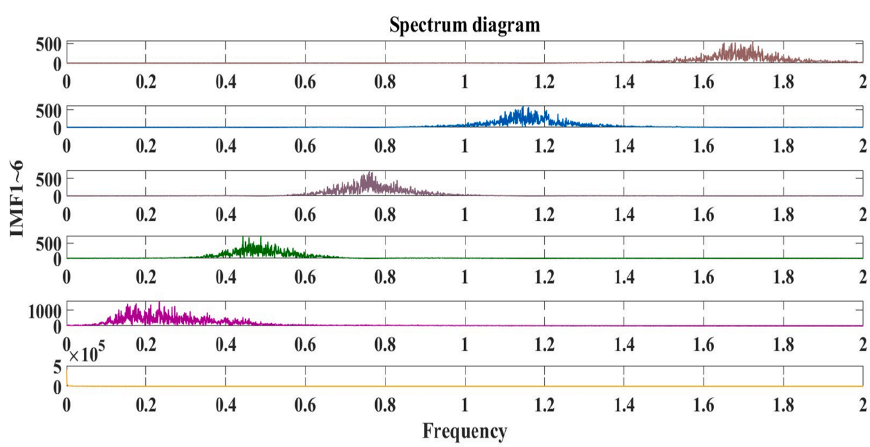


**Fig. 16.** VMD spectrum.

suggesting it fails to explain the data variability effectively. The BiLSTM model (MSE = 70.4116, $R^2$ = 76.171) offers a significant improvement over LSTM, demonstrating the value of bidirectional processing in capturing more accurate predictions. The GRU model (MSE = 22.59, $R^2$ = 83.7865) performs well, with a high $R^2$ and lower errors, outperforming LSTM, BiLSTM, SVM, RVM, CNN-SVM, and Transformer, but still lagging behind the CNN-BiLSTM and CEEMDAN-VMD-CNN-BiLSTM models. The SVM model achieves a moderate performance with an MSE of 40.0923 and $R^2$ of 82.5054, outperforming LSTM, BiLSTM, and Transformer, but still underperforming compared to CNN-BiLSTM and the proposed CEEMDAN-VMD-CNN-BiLSTM model. The Transformer model, with an MSE of 116.5105 and $R^2$ of 72.8989, ranks lower than most other models, including the CEEMDAN-VMD-CNN-BiLSTM model.

In summary, the CEEMDAN-VMD-CNN-BiLSTM model stands out with its superior predictive accuracy, significantly reducing prediction errors and achieving the highest $R^2$ value, thereby proving to be the most effective model for forecasting crude oil prices.

The CEEMDAN-VMD-CNN-BiLSTM model outperforms the other eight models in terms of prediction accuracy, with significantly lower error metrics and higher goodness-of-fit ($R^2$ values) compared to all eight benchmark models, including LSTM, BiLSTM, SVM, RVM, GRU, CNN-SVM, CNN-BiLSTM, and Transformer. This validates the effectiveness of the strategy combining signal decomposition techniques (CEEMDAN, VMD) with a deep learning hybrid model (CNN-BiLSTM) for enhancing performance in this forecasting task.

The LSTM, BiLSTM, SVM, RVM, GRU, CNN-SVM, CNN-BiLSTM, and Transformer models, though widely used for time-series forecasting, each have inherent limitations when applied to volatile and non-stationary datasets like crude oil prices. LSTM and BiLSTM struggle to capture long-term dependencies and complex nonlinear relationships, particularly in the presence of high noise and volatility. SVM and RVM are less effective in handling large-scale, non-linear time-series data and fail to capture the temporal dynamics between factors influencing price movements. GRU is an improvement over LSTM but still faces challenges in dealing with complex market fluctuations. CNN-SVM combines CNN's ability to extract features with SVM's regression capabilities but lacks the ability to handle long-term dependencies and time-series patterns effectively. CNN-BiLSTM is better at feature extraction and temporal modeling but still does not account for non-stationary or multi-frequency patterns in the data. The Transformer model, while

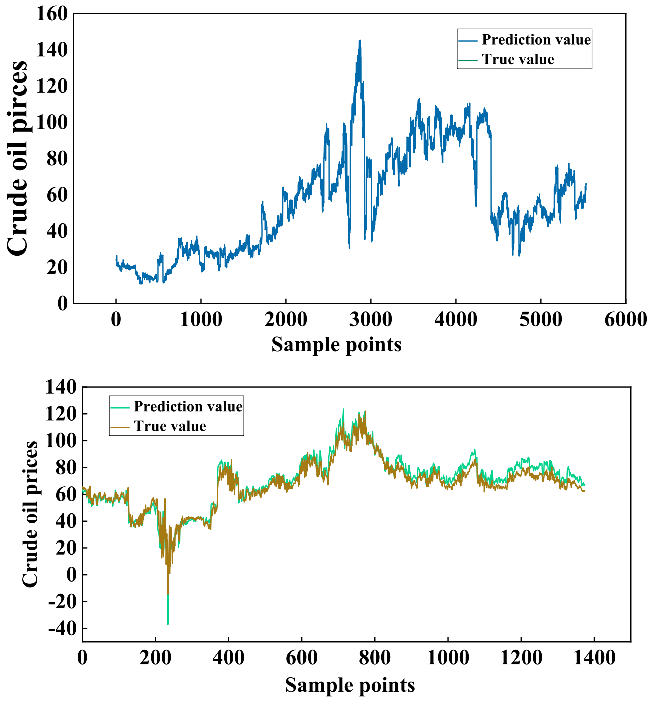


**Fig. 17.** Performance evaluation of the CEEMDAN-VMD-CNN-BiLSTM model on training and test sets.

**Table 5**
Significance test of the new model against other model.

| Model | BiLSTM | CNN-BiLSTM | VMD-CNN-BiLSTM |
|---|---|---|---|
| P-Values | $1.7 \times 10^{-28}$ | $2.23 \times 10^{-28}$ | 0.014524 |

**Table 6**
Prediction errors under different models.

| Model | Error Metrics | | | | |
|---|---|---|---|---|---|
| | MSE | RMSE | MAE | MAPE | $R^2$ |
| LSTM | 247.6859 | 15.738 | 14.532 | 21.6465 | 44.3165 |
| BiLSTM | 70.4116 | 8.3912 | 5.9572 | 11.5542 | 76.171 |
| SVM | 40.0923 | 6.3319 | 4.5818 | 7.4029 | 82.5054 |
| RVM | 173.7809 | 13.1826 | 8.6562 | 16.537 | 27.4661 |
| GRU | 22.59 | 4.7529 | 3.1698 | 6.2145 | 83.7865 |
| CNN-SVM | 71.0146 | 8.427 | 6.0706 | 10.7946 | 33.6605 |
| CNN-BiLSTM | 31.4605 | 5.609 | 4.0583 | 6.2542 | 90.528 |
| Transformer | 116.5105 | 10.794 | 7.8373 | 10.7994 | 72.8989 |
| CEEMDAN-VMD-CNN-BiLSTM | 24.168 | 4.9161 | 3.9259 | 5.5691 | 93.8742 |

effective in handling long-range dependencies, struggles with high-frequency volatility and noise in price data.

In contrast, the CEEMDAN-VMD-CNN-BiLSTM model overcomes these limitations by integrating CEEMDAN and VMD decomposition techniques, allowing it to effectively decompose complex crude oil price signals into high-frequency noise, mid-frequency cycles, and low-frequency trends. This multi-level decomposition improves model accuracy by capturing both short-term fluctuations and long-term trends. Furthermore, the CNN and BiLSTM combination excels at feature extraction and temporal modeling, enabling the model to learn intricate patterns across multiple time scales. This comprehensive approach leads to superior forecasting performance, especially during periods of high market uncertainty and volatility, setting the CEEMDAN-VMD-CNN-BiLSTM model apart from traditional methods.

As shown in Table 7, The CEEMDAN-VMD-CNN-BiLSTM model demonstrates exceptional performance, outperforming other models such as LSTM, BiLSTM, SVM, RVM, GRU, CNN-SVM, CNN-BiLSTM, Transformer, and VMD-CNN-BiLSTM, as indicated by significantly low p-values (ranging from $1.7 \times 10^{-117}$ to 0.014524). These p-values, all well below the threshold of 0.05, confirm the statistical significance of the model's superior prediction accuracy and robustness. The results highlight the model's ability to effectively handle complex, non-linear data and generalize across various datasets, making it a highly reliable and advanced choice for forecasting crude oil prices.

The proposed CEEMDAN-VMD-CNN-BiLSTM model is highly suitable for forecasting WTI crude oil prices due to its ability to effectively capture both short-term fluctuations and long-term trends in a volatile and non-stationary time series. This model integrates multiple advanced techniques to address the unique challenges associated with crude oil price forecasting, setting it apart from existing methods. The improvements and contributions of the proposed method compared to other models can be summarized as follows:

(1) Signal Decomposition: It uses CEEMDAN and VMD for effective decomposition, separating price series into high-frequency fluctuations and low-frequency trends, improving model accuracy by addressing nonlinearity and non-stationarity.
(2) Hybrid Deep Learning: The combination of CNN and BiLSTM allows the model to capture both short-term fluctuations and long-term trends. BiLSTM's bidirectional processing enhances the model's ability to learn from past and future dependencies.

In summary, the CEEMDAN-VMD-CNN-BiLSTM model improves forecasting by combining advanced signal decomposition with deep learning, resulting in more accurate, robust, and comprehensive predictions compared to existing models.

## 4. Conclusion

This study introduces the CEEMDAN-VMD-CNN-BiLSTM hybrid model to forecast WTI crude oil prices, addressing challenges like nonlinearity and non-stationarity. The model employs CEEMDAN and VMD for dual decomposition of price series, separating high-frequency noise, mid-frequency cycles, and low-frequency trends. CNN extracts local temporal features, while BiLSTM captures long- and short-term dependencies, improving prediction accuracy. The main findings are as follows:

(1) Through Pearson correlation analysis, four key factors were identified: LBMA gold price, Brent crude oil price, RMB-USD exchange rate, and federal funds rate, forming a reliable feature set for modeling.
(2) The hybrid model reduces the prediction MAPE to 5.57 %, a 40 % improvement over individual models. The synergy of CEEMDAN, VMD, CNN, and BiLSTM enhances forecasting performance.
(3) Hyperparameter optimization and ablation experiments confirmed the model's effectiveness, with CEEMDAN improving signal-to-noise ratio by 2.3 dB, VMD enhancing high-frequency component energy by 18 %, and CNN-BiLSTM reducing temporal modeling error by 27 %. The model achieves an $R^2$ of 93.87 %.

The CEEMDAN-VMD-CNN-BiLSTM model offers significant practical value, providing precise predictions for energy companies, financial markets, and policymakers to optimize strategies and manage risks.

### CRediT authorship contribution statement

**Shijie Zhu:** Writing – review & editing, Funding acquisition. **Mei Xu:** Writing – review & editing. **Jie Wu:** Formal analysis. **Yaning Wang:** Methodology, Investigation, Data curation, Conceptualization. **Xinsheng Jiang:** Methodology. **Zhuangzhuang Huang:** Validation,

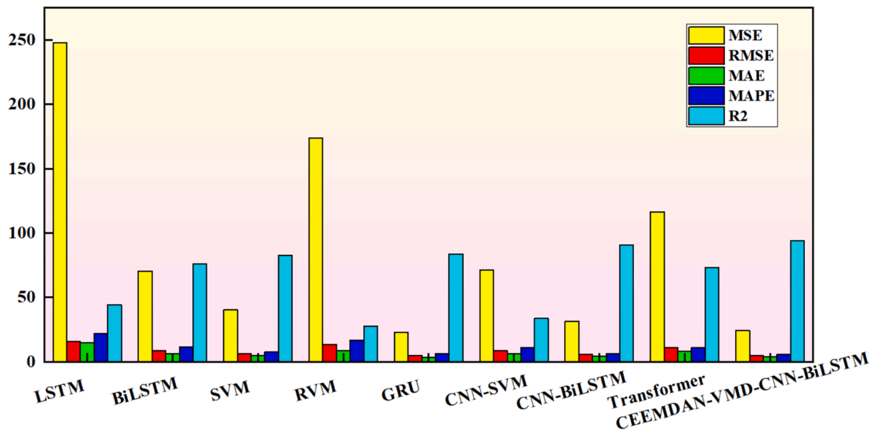


**Fig. 18.** Evaluation metrics for different models.

**Table 7**
Significance test of the new model against other model.

| LSTM | BiLSTM | SVM | RVM | GRU | CNN-SVM | CNN-BiLSTM | Transformer | VMD-CNN-BiLSTM |
|---|---|---|---|---|---|---|---|---|
| $1.7 \times 10^{-117}$ | $1.7 \times 10^{-28}$ | $9.61 \times 10^{-8}$ | $9.74 \times 10^{-41}$ | $2.34 \times 10^{-6}$ | $3.79 \times 10^{-80}$ | $2.23 \times 10^{-28}$ | $2.4 \times 10^{-25}$ | 0.014524 |

Supervision. **Yang Wang:** Validation. **Yong Zhu:** Validation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rineng.2025.106391.

## Appendix

Definitions of Different Abbreviations

| Abbreviation | Full Term |
|---|---|
| WTI | West Texas Intermediate |
| CEEMDAN | Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise |
| VMD | Variational Mode Decomposition |
| CNN | Convolutional Neural Network |
| BiLSTM | Bidirectional Long Short-Term Memory |
| MAPE | Mean Absolute Percentage Error |
| $R^2$ | Coefficient of Determination |
| IMF | Intrinsic Mode Function |
| SVM | Support Vector Machine |
| LSTM | Long Short-Term Memory |
| GRU | Gated Recurrent Unit |
| XGBoost | Extreme Gradient Boosting |
| EEMD | Ensemble Empirical Mode Decomposition |
| SVR | Support Vector Regression |
| ARIMA | Autoregressive Integrated Moving Average |
| MAE | Mean Absolute Error |
| RMSE | Root Mean Squared Error |
| MSE | Mean Squared Error |
| VIX | Volatility Index (Chicago Board Options Exchange Volatility Index) |
| LBMA | London Bullion Market Association |
| FRED | Federal Reserve Economic Data |
| ADMM | Alternating Direction Method of Multipliers |
| OVMD | Optimized Variational Mode Decomposition |
| ATT | Attention Mechanism |
| SSF | Smoothness-Scale Fusion |
| DBILSTM | Deep Bidirectional Long Short-Term Memory |

(continued on next page)

(*continued*)

| Abbreviation | Full Term |
|---|---|
| HMOFTTA | Hybrid Multi-Objective Fuzzy Time-Functioning Algorithm |
| VIKOR | Multi-criterion compromise solution ranking method |
| TOPSIS | **T**echnical for **O**rder of **P**reference by **S**imilarity to **I**deal **S**olution |
| RVM | Relevance Vector Machine |
| CNN-BiLSTM | Convolutional Neural Network - Bidirectional Long Short-Term Memory |
| VMD-CNN-BiLSTM | Variational Mode Decomposition - Convolutional Neural Network - Bidirectional Long Short-Term Memory |
| CEEMDAN-VMD-CNN-BiLSTM | Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise - Variational Mode Decomposition - Convolutional Neural Network - Bidirectional Long Short-Term Memory |

## Data availability

The data that has been used is confidential.

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